

MA 765: Topics in Non-linear Algebra

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1 Gröbner Basis

Denote by $\leq$ the coordinate wise partial order on $\mathbb{N}_0^n$. $(a_1, \dots, a_n) \leq (b_1, \dots, b_n)$ if $a_i \leq b_i$ for $i \in [n]$.

Divisibility is a partial order on monomials.

Theorem 1.1 (Dickson's lemma). Every infinite subset of $\mathbb{N}_0^n$ contains elements a, b with $a < b$.

Proof. The proof follows by induction. Let $\mathcal{M} \subset \mathbb{N}_0^n$ be an infinite subset. For any $i \in \mathbb{N}_0$ define $\mathcal{M}_i$,

$$\mathcal{M}_i = \{a = (a_1, \dots, a_n) \in \mathbb{N}_0^{n-1} : (a, i) \in \mathcal{M}\}$$

If $\mathcal{M}_i$ is finite, then look at $\bigcup_i \in \mathbb{N}_0$, which has finitely many and atleast one minimal element. Thus there is some $j \in \mathbb{N}$ such that $\bigcup_{i=0}^j \mathcal{M}_i$ with $a \in \mathcal{M}_i$ for some $i \leq j$. Hence $(a, i) < (b, k)$. □

Corollary 1.2. For any $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ the set of minimal elements wrt $<$ is nonempty and finite.

1.3. Monomial order on $\mathbb{N}_0^n$ is a relation $<$ such that

1. If $a \neq b$ then $a < b$ or $b < a$.
2. If $A < b$ and $b < c$ then $a < c$.
3. $(0, \dots, 0) < a$ for any $a \in \mathbb{N}_0^n$.
4. $a < b$ then $a + c < b + c$ for any $c \in \mathbb{N}_0^n$.

First two conditions together imply that $<$ is a total order.

Remark 1.4. 1. If $a < b$, then $a < b$, i.e., any monomial order refines coordinate wise partial order.

2. Any monomial order on $\mathbb{N}_0^n$ gives total order on monomials in $k[X_n]$

Example 1.5. Lexicographic order: Left most non zero entry of $a - b$ is positive the $a >_{lex} b$.

degree lex order if either $|a| > |b|$ or $|a| = |b|$ and right most entry of $a - b$ is negative.

degree reverse lex order, same as above but right most entry of $a - b$ is negative.

Proposition 1.6. For any monomial order $<$ on $\mathbb{N}_0^n$ with $\phi \neq \mathcal{M} \subset \mathbb{N}_0^n$ has a unique minmal element.

Proof. Dickson's lemma gives that $\mathcal{M}$ has a finite non empty set of minmal elements wrt coordinate wise order. The minmal elet of thse wrt $<$ is the desired elemnt. □

1.7. Fix a monomial order $<$ on $k[X_n]$

1. The initial monomial $\text{im}_{<}(f)$ of $f = \sum c_a x^a$ is the largest monomial wrt $<$ appearing in f with non zero coefficients.
2. The leading term $\text{lt}_{<}(f) := c_a x^a$ with $x^a = \text{im}(f)$

3. The initial ideal of an ideal I is $\text{im}(I) = \langle \text{im}(f) : f \in I \rangle$.

If G is a generating set for an ideal I then $\langle \text{im}(g) : g \in G \rangle \subset \text{im}(I)$ and this inclusion can be strict.

Proposition 1.8. Let $<$ be a monomial order on $k[X_n]$. Every ideal I has a finite subest $\mathcal{G}$ such that $\text{im}(I) = \langle \text{im}(g) : g \in \mathcal{G} \rangle$.

Any such $\mathcal{G}$ is called a Gröbner Basis of I wrt $<$.

Proof. The set of monomials in $\text{im}(I)$ has a finite and nonempty subset of minimal elements wrt divisibility, say $m_1, \dots, m_s$. Thus $\text{im}(I) = \langle m_1, \dots, m_s \rangle$. Every monomial in $\text{im}(I)$ is the initial monomial of some $f \in I$. hence there exists $f_1, \dots, f_s \in I$ with $\text{im}(f_i) = m_i$. Thus $\{f_i\}$ is a Gröbner basis. $\square$

Theorem 1.9. If $\mathcal{G}$ is a Gröbner basis of I , then $I = \langle \mathcal{G} \rangle$

Note that Hilbert's basis theorem is a simple corollary of this.

Proof. We argue by contradiction. By 1.6, choose $f \in I - \langle \mathcal{G} \rangle$ such that $\text{im}(f)$ is minimal. Call $\text{im}(f) = x^b$. $x^b \in \text{im}(I) = \langle \text{im}(g) : g \in \mathcal{G} \rangle$. There exists $g \in \mathcal{G}$ such that $\text{im}(g) | x^b$, say $x^b = x^c \cdot \text{im}(g)$.

$\text{im}(f - x^c \lambda g) < x^b = \text{im}(f)$ where $\lambda = \text{lt}(f)/x^c \text{lt}(g) \in k$. But $f - x^c \lambda g \in I - \langle \mathcal{G} \rangle$, which is a contradiction to the minimality. $\square$

Lemma 1.10. Consider $f, g_1, \dots, g_s \in k[X_n]$, with $g_i \neq 0$. Then for any minimal order $<$, there exists $q_1, \dots, q_s, r \in k[X_n]$ such that

1. $f + \sum_{i=1}^s q_i g_i = r$
2. $\text{im}(f) \geq \text{im}(q_i g_i) \forall i$ (note that $\text{im}(f) = \text{im}(q_i g_i)$ for some i)
3. $\text{im}(r)$ is not divisible by $\text{im}(g_i)$ for any i .

We say f reduces to r by $\{g_1, \dots, g_s\}$.

1.11 Division Algorithm. Input: $f, g_1, \dots, g_s$

Output: $q_1, \dots, q_s, r$ satisfying the properties 1-3.

1. Set $r = 0, p = gf, q_1 = \dots = q_s = 0$

2. While $p \neq 0$ do :

If $\text{im}(g_i)$ divides $\text{im}(p)$ for some $i \in [s]$, then set $q_i = q_i + \frac{\text{lt}(p)}{\text{lt}(g_i)}$ and $p_i = p - \frac{\text{lt}(p)}{\text{lt}(g_i)} g_i$
else set $r = r + \text{lt}(p), p = p - \text{lt}(p)$

3. Return $q_1, \dots, q_s, r$

Example 1.12. Consider Lexicographic ordering with $x > y$ and $f = x^2y + xy^2 + y^2$ and

$$g_1 = xy - 1, g_2 = y^2 - 1.$$

p	$xy - 1$	$y^2 - 1$	r
$x^2y + xy^2 + y^2 - x^2y + x$	x		
$xy^2 + y^2 + x - xy^2 + y$	y		
$y^2 + x + y - x$			x
$y^2 + y - y^2 + 1$	1		
$y + 1 - y$			y
1			1

Corollary 1.13. Buchberger's criteria Let $\mathcal{G}$ be a finite subset of I . Then $\mathcal{G}$ is a Gröbner basis of I iff each $f \in I$ can be reduced to 0 by $\mathcal{G}$.

Proof. If f reduces to r by $\mathcal{G}$, then $\text{im}(g_i)$ does not divide $\text{im}(r)$, for all i . However $f - r \in \langle \mathcal{G} \rangle$ and $r \in I$ and $\mathcal{G}$ is a Gröbner basis of I . So there exists some $g \in \mathcal{G}$ such that $\text{im}(r)$ is divisible by $\text{im}(g)$, which forces $r = 0$.

Conversely, we have $f = \sum q_i g_i$ with $g_i \in \mathcal{G}$ and $\text{im}(q_i g_i) \leq \text{im}(f)$. Hence equality for some i and so $\text{im}(f) \in \langle \text{im}(g) : g \in \mathcal{G} \rangle$. $\square$

Proposition 1.14. Let $<$ be a monomial order. Then

1. Let B be the set of monomials in $k[X_n] - \text{im}(I)$. Then $\bar{B} \subset k[X_n]/I$ is a k vsp basis.
2. If $\mathcal{G}$ is a Gröbner basis of I , then the remainder of f by $\mathcal{G}$ is unique and does not depend on the choice of $\mathcal{G}$.

Proof. 1. If $p = \sum \lambda_i m_i \in I$ with $m_i \in B$, then $\text{im}(p) \in \text{im}(I)$, but $\text{im}(p) = \text{im}(m_i) \notin I$. To show $\bar{B}$ spans $m \in k[X_n]$ such that $\bar{m} \notin \text{span}(\bar{B})$.

Take $\min\{m\} = m$ where $m \notin B$. So we have $m \in \text{im}(I)$. There exists $f \in I$ such that $\text{im}(f) = m$. So any monomial in $f - \text{lt}(f) + I = f - \lambda m + I$ is in $\text{span}(\bar{B})$. So $\lambda m + I = p - f + I \in \text{span}(\bar{B})$. This leads to a contradiction $\square$

Definition 1.15. 1. For terms $\lambda x^a, \mu x^b$ ($\lambda, \mu \in k$) denote by

$$\begin{aligned} \gcd(\lambda x^a, \mu x^b) &= \gcd(x^a, x^b) \\ \text{lcm}(\lambda x^a, \mu x^b) &= \text{lcm}(x^a, x^b) \end{aligned}$$

2. For $0 \neq g, h \in k[x_n]$, their s -polynomial (wrt monomial order $<$)

$$S(g, h) := \frac{\text{lt}(h)}{\gcd(\text{lt}(h), \text{lt}(g))} g - \frac{\text{lt}(g)}{\gcd(\text{lt}(h), \text{lt}(g))} h.$$

1.16. Buchberger's Algorithm for computing Gröbner basis Input: $f_1, \dots, f_s \in k[X_n]$ in monomial order.

Output: Gröbner basis $\mathcal{G}$ of $I = \langle f_1, \dots, f_s \rangle$ wrt $<$.

1. Set $\eta = \langle f_1 \dots, f_s \rangle$
2. Order the elements of $\mathcal{G}$ as $f_1, \dots, f_t$
3. For $1 \leq i < j \leq t$ do: Reduce $S(g_i, g_j)$ to r by g . If $r \neq 0$, then set $\mathcal{G} := \mathcal{G} \cup \{r\}$ and go to step 2.
4. Return $\mathcal{G}$

Remark 1.17. The algorithm computes a Gröbner basis. It terminate because in case $r \neq 0$, $\text{im}(r) \notin \langle \text{im}(g_1), \dots, \text{im}(g_t) \rangle$.

1.18. Extension to submodules of finitely generated $k[X_n]$ module So $F = k[X_n]^r = \bigoplus_{i=1}^r k[X_n]e_i$.

Monomials in G are of the form $x^a e_i$ and terms are $\lambda x^a e_i$ with $\lambda \in k$.

A monomial order on FF is a total order on the monomials satisfying:

If $x^a \neq 1$, then $m_1 < m_2 \implies m_1 < x^a \cdot 1 < x^a m_2$, for monomials m_i in F .

Given a monomial order on the polynomial ring $k[X_n]$ and an order on $\{e_i\}$. We obtain a monomial ordering on F by ordering $\mathbb{N}_0^n \times [r]$ or $[r] \times \mathbb{N}_0^n$ lexicographically.

Dicksen's lemma can be extended to monomials in F . We also define im , It with respect to $<$ analogously. There is also a division algorithm.

Theorem 1.19. 1. Every submodule M of F has finite Gröbner basis and the basis generates M .

2. If B is the set of monomials in $F - \text{im}(M)$, then $\bar{B} \subset F/M$ form a k basis of F/M .

2 Hilbert Functions

Definition 2.1. Let $I \subset k[X_n]$ be a monomial ideal. The Hilbert function of $A = k[X_n]/I$ (or of I) is

$$h_A : \mathbb{N}_0 \rightarrow \mathbb{Z}$$

$$a \mapsto h_A(j) = \dim_k[A]_j$$

where $[A]_j$ is k vector space of images of polynomials of degree j , from $k[X_n] \rightarrow A$. (So $h_A(j) = \text{number of monomials in } [k[X_n]]_j - I$).

It's generating function is the Hilbert series

$$H_A(z) = \sum_{j \geq 0} h_A(j) z^j$$

Example 2.2. 1. For $I = 0$, we get $h_{k[X_n]}(j) = \binom{n+j-1}{j}$ and so

$$H_{k[X_n]}(z) = \sum_{j \geq 0} \binom{n+j-1}{j} z^j = \frac{1}{(1-z)^n}$$

2. If $I = \langle x^a \rangle$, then let $e = \deg(x^a)$.

3.

$$h_{k[X_n]/I}(j) = \begin{cases} h_{k[X_n]}(j) & j < e \\ h_{k[X_n]}(j) - h_{k[X_n]}(j - e) & j \geq e \end{cases}$$

Hence

$$H_{k[X_n]/I} = \frac{1 - z^e}{(1 - z)^n}$$

Theorem 2.3. For any proper monomial ideal I of $k[X_n]$, the Hilbert series of $A = k[X_n]/I$ is a rational function of the form

$$H_A(z) = \frac{\kappa_A(z)}{(1 - z)^d}$$

with $\kappa_A(z) \in \mathbb{Z}[z]$, $\kappa_A(0) = 1$, $\kappa_A(1) \neq 0$, $d \in \mathbb{N}$. (this expression is unique.)

The dimension of A is

$$\dim A := d$$

and the multiplicity of A (or degree of I) is

$$\deg(I) = \mathcal{H}_A(1) > 0$$

There is a polynomial called Hilbert polynomial p_A of A such that $h_A(j) = p_A(j)$ if $j \gg 0$. If $d = \dim A > 0$, then

$$\begin{aligned} p_A(z) &= \frac{\deg(I)}{(d-1)!} z^{d-1} + \text{lower order terms} \\ &= h_0(A) \binom{z+d-1}{d-1} + h_1(A) \binom{z+d-2}{d-2} + \dots + h_{d-1}(A) \binom{z}{0} \end{aligned}$$

where $h_0(A) = \deg(I)$ and $h_i(A)$ are integers. Note if $p_A(z) \neq 0$ polynomial, then $\deg p_A = d - 1 = \dim A - 1$.

Example 2.4.

$I = 0$, we get $\dim k[X_n] = n$ and the multiplicity of $k[X_n]$ is $1 = \deg I$.

$I = \langle x^a \rangle$ with $\deg x = e$, then $\dim k[X_n]/I = n - 1$ and $\deg(I) = \deg(x^a)$.

Proof of 2.3. Inclusion exclusion principle: $|\bigcup_{i=1}^s X_i| = \sum_{\phi \neq T \subset [s]} (-1)^{|T|+1} |X_T|$ where $X_T = \bigcap_{i \in T} X_i$

Let $I = \langle m_1, \dots, m_s \rangle = I$, where m_i are monomials. Denote by $X_i(j)$ set of degree j monomials in $\langle m_i \rangle \subset k[X_n]$. hence for $T \subset [s]$, $X_T(j) = \bigcap_{i \in T} X_i(j)$ is the set of deg j monomials that are divisible by $m_T = \text{lcm}(m_i)_{i \in T}$. Define $e_t := \deg m_T$. So

$$|X_T(j)| = \begin{cases} 0 & j < e_t \\ \binom{n-1+j-e_t}{n-1} & j \geq e_t \end{cases}$$

Thus

$$\sum_{j \geq 0} |X_T(j)| z^j = \sum_{j \geq e_t} \binom{n-1+j-e_t}{n-1} z^j = \sum_{k \geq 0} \binom{n-1+k}{n-1} z^{k+e_t} = z^{e_t} \frac{1}{(1-z)^n}$$

Since

$$\begin{aligned} h_A(j) &= \binom{n-1+j}{n-1} - \text{number of deg } j \text{ monomials in } I = \langle m_1 \dots, m_s \rangle \\ &= \binom{n-1+j}{n-1} - \left| \bigcup_{i \in [j]} X_i \right| \end{aligned}$$

We get

$$\begin{aligned} H_A(z) &= \sum_{j \geq 0} h_A(j) z^j = \sum_{j \geq 0} \binom{n-1+j}{n-1} z^j + \sum_{T \in [s]} (-1)^{|T|} \sum_{j \geq 0} |X_T(j)| z^j \\ &= \frac{1}{(1-z)^n} + \sum_{T \in [s]} (-1)^T \frac{z^{e_T}}{(1-z)^n} =: \frac{g(z)}{(1-z)^n} \end{aligned}$$

When $g(z) \in \mathbb{Z}[z]$, with $g(0) = 1$ writing $g(z) = (1-z)^v \kappa_A(z)$ with $\kappa_A(z) \in \mathbb{Z}[z]$, suitable $v \in \mathbb{N}$ and $\kappa_A(1) \neq 0, \kappa_A(0) = 1$. Hence $H_A(z) = \frac{\kappa_A(z)}{(1-z)^d}$ where $d = n - v$. Write

$$\kappa_A(z) = \sum_{k=0}^w c_k z^k \text{ with } c_k \in \mathbb{Z}$$

$$\sum_{j \geq 0} h_A(j) z^j = \left(\sum_{k=0}^w c_k z^k \right) \left(\sum_{l \geq 0} \binom{d-1+l}{d-1} z^l \right)$$

comparing coefficients in $\deg j \gg 0$, we get

$$\begin{aligned} h_A(j) &= \sum_{k+l=j} c_k \binom{d-1+l}{d-1} = \sum_{k=0}^w c_k \underbrace{\binom{d-1+j-k}{d-1}}_{\text{polynomial in } j-k \text{ variables of deg } d-1} \\ &= \underbrace{\left(\sum_{k=0}^w c_k \right)}_{\kappa_A(1)} \binom{j}{d-1} + \text{lower order terms} \\ &=: p_A(j) \text{ (Hilbert polynomial)} \end{aligned}$$

If $h_A(j) = 0$ whenever $j \gg 0$, then by definition $0 = d = \dim A$ and in this case p_A is the zero polynomial. Hence if $d > 0$, then $h_A(j) > 0$ if $j \gg 0$ and so the leading coefficient of $p_A(z)$ must be positive, i.e., $\kappa_A(1) > 0$ $\square$

Definition 2.5. A monomial order is called degree compatible if

$$\deg(x^a) > \deg(x^b) \implies x^a > x^b$$

for any two monomials.

2.6. For any ideal $I \subset k[X_n]$ and any $t \in \mathbb{Z}$, set

$$I_{\leq t} = \{f \in I : \deg f \leq t\}$$

It is a k -subspace of $k[X_n]$. Write $\text{Mon}(k[X_n])$ for the set of monomials in $k[X_n]$.

Lemma 2.7. Let $<$ be a degree compatible monomial order. For any ideal $I \subset k[X_n] = S$, one has

$$\dim_k \frac{k[X_n]_{\leq t}}{I_{\leq t}} = \text{number of monomials in } \frac{k[X_n]_{\leq t}}{\text{im}_{<}(I)} \\ = | \text{Mon}(k[X_n])_{\leq t} - \text{im}_{<}(t) |$$

Proof. $B := \text{Mon}(k[X_n])_{\leq t} - \text{im}_{<}(I)$. We claim

$$\bar{B} \subset \frac{k[X_n]_{\leq t}}{I_{\leq t}} \text{ is a } k\text{-basis}$$

$\overline{\text{Mon}(k[X_n]) - \text{im}(I)}$ is a k -basis of $k[X_n] - \text{im}(I)$, by 1.14. $\bar{B}$ spans remainder of any $F \in k[X_n]$ upon dividing by Gröbner basis of I wrt $<$ satisfies $\deg r \leq \deg f$. $\square$

For $I \subset k[X_n]$, the affine Hilbert function of $A = k[X_n]/I$ is

$$h_A^a : \mathbb{N}_0 \longrightarrow \mathbb{Z} \\ j \mapsto h_A^a(j) = \dim_k \frac{k[X_n]_{\leq j}}{I_{\leq j}}$$

Lemma 2.8. For any degree compatible monomial order $<$ on I , one has

$$h_{\frac{k[X_n]}{\text{im}_{<}(I)}}(j) = h_A^a(j) - h_A^1(j-1)$$

where $A = k[X_n]/I$.

Proof. By definition, we have

$$h_{\frac{k[X_n]}{\text{im}(I)}}(j) = | \text{Mon}(k[X_n])_j - \text{im}(I) |$$

$$2.7 \implies h_A^a(j) = | \text{Mon}(k[X_n])_{\leq j} - \text{im}(I) |$$

$\square$

Remark 2.9. If $A \neq 0$, $h_A^a(0) = 1$. Then by 2.8 we have $h_A^a(j) = \sum_{k=0}^j h_{k[X_n]/\text{im}(I)}(k)$. It follows that generating function of h_A^a and $h_{\frac{k[X_n]}{\text{im}_{<}(I)}}$ have analogous properties

2.10. We define dimension of A ,

$$\dim \frac{k[X_n]}{I} := \dim_k \frac{k[X_n]}{\text{im}(I)}$$

and the degree

$$\deg(I) = \deg(\text{im}(I))$$

where $<$ is a degree compatible monomial order.

2.11. Let $G = (G, +)$ be an abelian group. A G -graded ring R is a family of subgroups $([R]_{a \in G}) \leq (R, +)$ such that

1. $R = \oplus_{a \in G} [R]_a$ (as $\mathbb{Z}$ -modules)

$$2. [R]_a \cdot [R]_b \subset [R]_{a+b}$$

The elements of $[R]_a$ are called homogeneous of degree a .

Example 2.12. Fine or $\mathbb{Z}$ -grading of $k[X_n]$, where $G = \mathbb{Z}^n$

$$[S]_a = \begin{cases} 0 & \text{if some } a_i < 0 \\ \{\lambda x^a : \lambda \in k\} & \end{cases}$$

Definition 2.13. $R = G$ -graded ring

1. G -graded R -module is an R -module M with a decomposition $([M]_a)_{a \in G}$ such that $M = \bigoplus_{a \in G} [M]_a$ and $[R]_a \cdot [M]_b \subset [M]_{a+b}$.
2. A G -graded or simply graded submoudle of such a graded M is a graded submodule $N \subset M$ such that

$$[N]_a \subset [M]_a$$

Lemma 2.14. For an arbitrary submodule N of a G graded R -module M the following are equivalent

1. N is a graded submodule
2. N has a generating set consisting of homogeneous elemnents
3. If $m = \sum_{a \in G} m_a$ with $m_a \in [M]_a$, then $m \in N$ iff each $m_a \in N$.
4. M/N is a G -graded R -module with grading $[M/N]_a := \frac{[M]_a + N}{N}$.

Proof.

□

Example 2.15. 1. M and N are G -graded, then so is $M \oplus N$ with grading $[M \oplus N]_a := [M]_a \oplus [N]_a$ (as R modules). So if R is G graded, then so is R^n .

2. $I \subset k[X_n]$ is a $\mathbb{Z}^n$ graded submodule if I is a monomial ideal.

3. A $\mathbb{Z}$ -graded or homogeneous ideal of $k[X_n]$, is an ideal that has generating set consisting of homogeneous polyonomials. In that case $k[X_n]/I$ is a graded module.

2.16. A homomorphism of G -graded modules or a G -graded homomorphism is a R -module homomorphism $\phi : M \rightarrow N$ that is degree preserving, $\phi([M]_a) \subset [N]_a$.

For any $a \in G$ and a G -graded module M , the module $M(a)$ has the same module structure as M , but grading given by

$$[M(a)]_b := [M]_{a+b}$$

$M(a)$ is a degree a shift of M . (Note here that the convention is opposite of that in algebraic topology.)

Example 2.17. 1. Consider $k[X_n]$ with standard grading.

$$\begin{aligned} \phi : k[X_n] &\rightarrow k[X_n] \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned}\psi &: k[X_n](-2) \rightarrow k[X_n] \\ f &\mapsto x_1^2 f\end{aligned}$$

then $f \in k[X_n](-2)$ has degree $\deg f + 2$. So $f \in [k[X_n](-2)]_{\deg f + 2}$.

2. For any $a \neq \text{in } G$, $R(a)$ is not a graded ring, (because identity is not in 0 dimension), but it is a graded R -module.

Lemma 2.18. If $\phi : M \rightarrow N$ is a homomorphism of graded modules, then $\ker \phi$, $\text{im} \phi$, $\text{coker} \phi$ are graded modules.

Proof. $[\ker \phi]_a = \ker \phi \cap [M]_a$, and $[\text{im} \phi]_a = \text{im} \phi \cap [N]_a$.

□

Example 2.19. 1. If M is a $\mathbb{Z}$ graded module with generators $m_1, \dots, m_t$ where $d_i = \deg m_i$, then

$$\begin{aligned}\phi &: \bigoplus_{i=1}^t R(-d_i) \longrightarrow M \\ \begin{bmatrix} r_1 \\ \vdots \\ r_t \end{bmatrix} &\mapsto \sum_{i=1}^t r_i m_i\end{aligned}$$

is a homomorphism of graded R -modules and is surjective.

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x, y] + S$ with standard grading.

$$\begin{aligned}\phi &: S(-3) \oplus S(-2) \oplus S(-4) \rightarrow I \\ \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} &\mapsto f_1 x^3 + f_2 xy + f_3 y^4\end{aligned}$$

$$\ker \phi : \left\langle \begin{bmatrix} y \\ -x^2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ y^3 \\ -x \end{bmatrix} \right\rangle \xleftarrow[\text{graded hom}]{\simeq} S(-4) \oplus S(-5)$$

There exists an exact sequence,

$$\begin{array}{ccccccc} & & & S(-3) & & & \\ & & & \oplus & & & \\ 0 \rightarrow & \begin{matrix} S(-4) \\ S(-5) \end{matrix} \oplus & \xrightarrow{\begin{bmatrix} y & 0 \\ -x^2 & y^3 \\ 0 & -x \end{bmatrix}} & \begin{matrix} S(-2) \\ \oplus \\ S(-4) \end{matrix} & \xrightarrow{\begin{bmatrix} x^3 & xy & y^4 \end{bmatrix}} & S & \rightarrow S/I \rightarrow 0 \end{array}$$

Definition 2.20. For any $\mathbb{Z}$ graded module M over $k[X_n]$ its Hilbert function is

$$\begin{aligned}h_m &: \mathbb{Z} \rightarrow \mathbb{Z} \\ j &\mapsto h_m(j) := \dim_k [M]_j\end{aligned}$$

assuming $[M]_j$ is finitely generated for all j .

Remark 2.21. For any monomial ideal $I \subset k[X_n]$, 2.1 and 2.20 agree.

$$\dim_k[k[X_n]/I]_j = |\text{Mon}(k[X_n])_j - I|$$

Proposition 2.22. For every graded submodule M of a finitely generated free $k[X_n]$ module F and any monomial order $<$ of F ,

$$h_{F/M}(j) = \frac{h_F(j)}{\text{im}(M)} \quad \forall j \in \mathbb{Z}$$

Corollary 2.23. For any finitely generated $k[X_n]$ submodule $m \neq 0$, Hilbert series is of the form

$$\begin{aligned} H_M(z) &:= \sum_{j \in \mathbb{Z}} h_M(j) z^j \\ &= \frac{\kappa_M(z) z^t}{(1-z)^d} \end{aligned}$$

where $\kappa_M(z) \in \mathbb{Z}[z]$, $\kappa_M(0) \neq 0$, $\kappa_M(1) > 0$, $t \in \mathbb{Z}$.

There is a Hilbert polynomial $p_M(z) \in \mathbb{Q}(z)$ such that

$$h_M(j) = p_M(j) \quad j \gg 0$$

(krull) dimension of M is defined as

$$\dim M = d$$

and the degree is

$$\deg(M) = \kappa_M(1)$$

Proof. Let M be generated by $m_1, \dots, m_t$ of degree $d_1, \dots, d_t$ respectively. Define

$$\begin{aligned} \phi : \bigoplus_{i=1}^t S(-d_i) &\twoheadrightarrow M \\ f &\mapsto x_1^2 f \end{aligned}$$

is not a graded homomorphism. However define

$$\begin{aligned} \psi : k[X_n](-2) &\rightarrow k[X_n] \\ \begin{bmatrix} f_1 \\ \vdots \\ f_t \end{bmatrix} &\mapsto \sum f_i m_i \end{aligned}$$

Set $N = \ker \phi$ and we have a short exact sequence $0 \rightarrow N \rightarrow F \rightarrow M \rightarrow 0$ and by rank nullity theorem we have $h_M(j) = h_F(j) - h_N(j)$. S and F have desired Hilbert series which are rational functions. So it is enough to show the same for N , which is same as $h_{F/\text{im}(N)}$.

$\text{im}(N)$ is generated by monomials. So

$$\frac{F}{\text{im}(N)} \cong \bigoplus_{i=1}^t \underbrace{\left(\frac{S}{J_i} \right)}_{\text{has the desired properties}} (-d_i)$$

for monomial ideals J_i . □

Example 2.24. 1. $S = k[X_n], F = \bigoplus_{i=1}^t S(-d_i)$. Then

$$H_{S(-d_i)}(z) = \frac{z_i^d}{(1-z)^n}$$

2. Consider $I = \langle x^3, xy, y^4 \rangle \subset k[x, y] + S$ with standard grading. $A = S/I$. We have the exact sequence,

$$\begin{array}{ccccccc} & & S(-3) & & & & \\ & & \oplus & & & & \\ S(-4) & & & & & & \\ \oplus & \rightarrow & S(-2) & \xrightarrow{\phi} & S & \rightarrow & A = S/I \rightarrow 0 \\ S(-5) & & \oplus & & [x^3 & xy & y^4] \\ & & S(-4) & & & & \end{array}$$

$$\begin{aligned} H_A(z) &= H_S(z) - H_{S(-3) \oplus S(-2) \oplus S(-4)}(z) + H_{S(-4) \oplus S(-5)}(z) \\ &= H_S(z) - H_{S(-3)}(z) - H_{S(-2)}(z) + H_{S(-4)}(z) + H_{S(-5)}(z) - H_{S(-4)}(z) \\ &= \frac{1 - z^3 - z^2 + z^5}{(1-z)^2} \\ &= 1 + 3z + 2z^2 + z^3 \end{aligned}$$

So $\dim A = 0$ and $\deg M = 6$ (polynomial evaluated at 1).

Lemma 2.25. Let $d \in \mathbb{N}$. Every $a \in \mathbb{N}$ admits a unique presentation of the form

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

with integers $k_d > k_{d-1} \cdots > k_s$. It is called the d -Macaulay presentation of a .

Example 2.26. For $d = 3$ and $a = 12$, we have

$$12 = \binom{5}{3} + \binom{2}{2} + \binom{1}{1}$$

Definition 2.27. If $a > 0$ with d -Macaulay presentation

$$a = \binom{k_d}{d} + \binom{k_{d-1}}{d-1} + \cdots + \binom{k_s}{s}$$

set

$$a^{(d)} = \binom{k_{d+1}}{d+1} + \binom{k_d}{d} + \cdots + \binom{k_{s+1}}{s+1}$$

Example 2.28. $12^{(3)} = 17$

Theorem 2.29 (Macaulay). Let $h : \mathbb{N}_0 \rightarrow \mathbb{Z}$, the following are equivalent,

1. There is some $n \in \mathbb{N}_0$ and some homogeneous ideal $I \subset k[X_n]$ such that Hilbert function of $A = k[X_n]/I$ is h .

2. There is a monomial ideal (Lexicographic ideal) $I \subset k[X_n]$ with $n = h(1)$ such that Hilbert function of A is h .
3. $h(0) = 1$ and

$$h(j+1) \leq h(j)^{\langle j \rangle} \text{ if } j > 0$$

Moreover for every graded k -algebra A , one has

$$h_A(j+1) = h_A(j)^{\langle j \rangle} \text{ if } j \gg 0$$

Example 2.30.

j	0	1	2	3	4	5	6
$\tilde{h}$	1	4	10	12	18	18	...
h	1	4	10	12	17	17	...

$\tilde{h}$ is not a possible Hilbert function because $12^{\langle 3 \rangle} = 17$. While h is a possible Hilbert function.